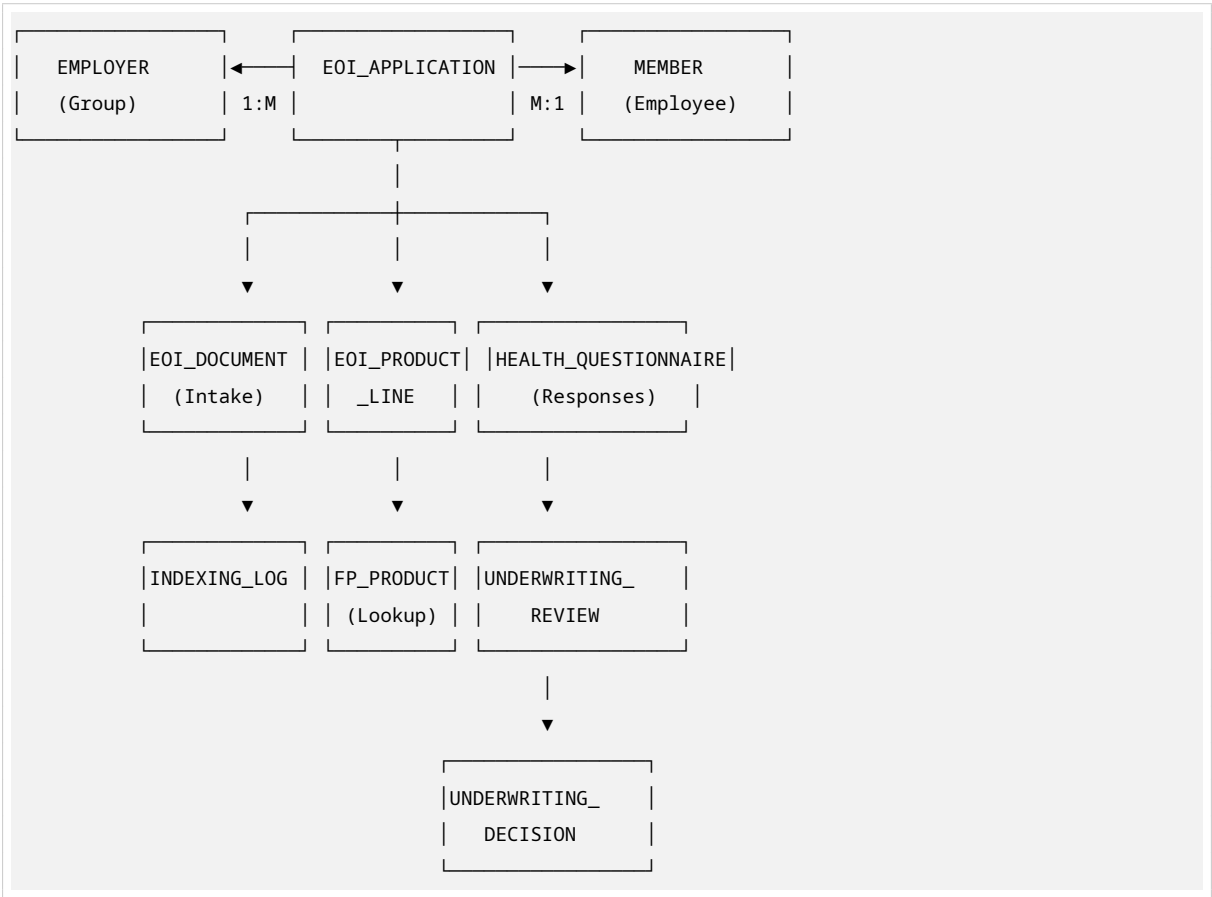


EOI Intake Process — Normalized Data Model

Overview

This document defines the normalized entity-relationship data model for the Evidence of Insurability (EOI) Intake Process. The model is designed in **Third Normal Form (3NF)** to eliminate redundancy while maintaining referential integrity across the Financial Protection (FP) workflow.

Entity Relationship Diagram (High-Level)



1. EMPLOYER (Group/Plan Sponsor)

Description: Commercial employers who contract with UHC to offer FP products to their employees.

Column Name	Data Type	Constraints	Description
employer_id	UUID	PK	Unique identifier for the employer
group_number	VARCHAR(50)	UNIQUE, NOT NULL	UHC-assigned group/plan number
employer_name	VARCHAR(255)	NOT NULL	Legal name of the employer
tax_id	VARCHAR(20)	UNIQUE	Employer Tax ID (EIN)

Table 1 – continued

Column Name	Data Type	Constraints	Description
industry_code	VARCHAR(10)	FK	SIC/NAICS industry classification
group_size	INT		Number of enrolled employees
effective_date	DATE	NOT NULL	Plan effective date
termination_date	DATE		Plan termination date (nullable)
is_active	BOOLEAN	DEFAULT TRUE	Whether the group plan is active
fp_products_offered	VARCHAR(500)		Comma-separated or JSON array of FP product codes offered
address_line_1	VARCHAR(255)		Employer mailing address
address_line_2	VARCHAR(255)		
city	VARCHAR(100)		
state	CHAR(2)		
zip_code	VARCHAR(10)		US state code
country	CHAR(2)	DEFAULT 'US'	
contact_name	VARCHAR(255)		
contact_email	VARCHAR(255)		
contact_phone	VARCHAR(20)		ISO country code
created_at	TIMESTAMP	DEFAULT NOW()	
updated_at	TIMESTAMP	DEFAULT NOW()	

Normalization: Employer data is separated from applications because one employer has many applications over time. Storing employer details here avoids repetition in every EOI record.

2. MEMBER (Employee/Applicant)

Description: Individual employees or dependents applying for FP coverage. Separated from application to support multiple EOI submissions over time.

Column Name	Data Type	Constraints	Description
member_id	UUID	PK	Unique member identifier
ssn	VARCHAR(11)	UNIQUE, NOT NULL	Social Security Number (encrypted at rest)
first_name	VARCHAR(100)	NOT NULL	Legal first name
middle_name	VARCHAR(100)		Middle name/initial
last_name	VARCHAR(100)	NOT NULL	Legal last name
date_of_birth	DATE	NOT NULL	DOB (MM/DD/YYYY)
gender	CHAR(1)		'M', 'F', 'N', 'U'

Table 2 – continued

Column Name	Data Type	Constraints	Description
height_inches	INT		Height in inches
weight_lbs	INT		Weight in pounds
marital_status	VARCHAR(20)		Single, Married, Domestic Partner, etc.
email	VARCHAR(255)		Personal email address
phone	VARCHAR(20)		Primary phone number
address_line_1	VARCHAR(255)		Residential address
address_line_2	VARCHAR(255)		
city	VARCHAR(100)		
state	CHAR(2)		
zip_code	VARCHAR(10)		
country	CHAR(2)	DEFAULT 'US'	
tobacco_use	BOOLEAN		Tobacco usage status
created_at	TIMESTAMP	DEFAULT NOW()	
updated_at	TIMESTAMP	DEFAULT NOW()	

Normalization: Member demographics are stored once and referenced by multiple EOI applications. This prevents storing the same SSN, DOB, and address repeatedly for every application.

3. FP_PRODUCT (Lookup/Master Data)

Description: Master reference table for all Financial Protection products offered by UHC.

Column Name	Data Type	Constraints	Description
product_code	VARCHAR(20)	PK	Unique product code (e.g., 'SUPP_LIFE', 'STD', 'LTD', 'CRIT_ILL')
product_name	VARCHAR(100)	NOT NULL	Human-readable product name
product_category	VARCHAR(50)	NOT NULL	'Life', 'Disability', 'Supplemental Health'
requires_eoi	BOOLEAN	DEFAULT TRUE	Whether EOI is required for this product
guaranteed_issue_amount	DECIMAL(12,2)		Max coverage without EOI (e.g., 200000.00)
min_coverage_amount	DECIMAL(12,2)		Minimum electable amount
max_coverage_amount	DECIMAL(12,2)		Maximum electable amount
is_active	BOOLEAN	DEFAULT TRUE	Whether product is currently offered
created_at	TIMESTAMP	DEFAULT NOW()	

Products:

product_code	product_name	category	requires_eoi
BASIC_LIFE	Basic Life Insurance	Life	FALSE
SUPP_LIFE	Supplemental Life Insurance	Life	TRUE
AD&D	Accidental Death & Dismemberment	Life	FALSE
STD	Short-Term Disability	Disability	TRUE
LTD	Long-Term Disability	Disability	TRUE
CRIT_ILL	Critical Illness	Supplemental Health	TRUE
ACCIDENT	Accident Insurance	Supplemental Health	FALSE
HOSP_INDEM	Hospital Indemnity	Supplemental Health	FALSE

Normalization: Product definitions are centralized. If guaranteed issue amounts change, update one row instead of every historical record.

4. EOI_APPLICATION (Core Intake Record)

Description: The central entity representing a single EOI submission. Links employer, member, and intake metadata.

Column Name	Data Type	Constraints	Description
application_id	UUID	PK	Unique application identifier
application_number	VARCHAR(50)	UNIQUE, NOT NULL	Business-facing application number (e.g., EOI-2026-00012345)
employer_id	UUID	FK → EMPLOYER	Group/plan sponsor reference
member_id	UUID	FK → MEMBER	Applicant reference
application_date	DATE	NOT NULL	Date the application was submitted
enrollment_type	VARCHAR(30)	NOT NULL	‘Initial’, ‘Open Enrollment’, ‘Late Entrant’, ‘Life Event’, ‘Increase’
intake_channel	VARCHAR(30)	NOT NULL	‘Paper’, ‘Portal_Adobe’, ‘Email’, ‘Fax’, ‘Census’
application_status	VARCHAR(30)	DEFAULT ‘Received’	‘Received’, ‘Indexing’, ‘Under Review’, ‘Pending Info’, ‘Approved’, ‘Denied’, ‘Withdrawn’
is_census	BOOLEAN	DEFAULT FALSE	Whether part of a batch census submission

Table 5 – continued

Column Name	Data Type	Constraints	Description
census_batch_id	UUID	FK → CENSUS_BATCH	Nullable reference to batch (future)
requested_effective_date	DATE		When coverage should begin
submitted_by	VARCHAR(100)		Name of person who submitted (if not member)
submitted_by_relationship	VARCHAR(50)		‘Self’, ‘HR’, ‘Broker’, ‘Spouse’
adobe_agreement_id	VARCHAR(100)		Adobe Sign agreement ID (if portal submission)
is_signature_valid	BOOLEAN		Whether signature passed validation
signature_date	TIMESTAMP		Date/time of signature
signature_type	VARCHAR(20)		‘Wet’, ‘Digital’, ‘E-Sign’
indexing_completed_at	TIMESTAMP		When indexing was finished
indexing_completed_by	VARCHAR(100)		User ID of indexer or ‘SYSTEM’ if automated
underwriting_assigned_at	TIMESTAMP		When assigned to underwriter
underwriter_id	VARCHAR(50)		ID of assigned underwriter
review_started_at	TIMESTAMP		When underwriter began review
review_completed_at	TIMESTAMP		When decision was rendered
turnaround_days	INT		Calculated: review_completed - application_date
created_at	TIMESTAMP	DEFAULT NOW()	
updated_at	TIMESTAMP	DEFAULT NOW()	

Normalization: Application is the fact table. It contains only foreign keys to employer and member, plus intake-specific attributes. No repeating groups.

5. EOI_PRODUCT_LINE (Junction: Application × Products)

Description: Many-to-many junction between applications and products. One EOI application can request multiple FP products.

Column Name	Data Type	Constraints	Description
product_line_id	UUID	PK	Unique line item ID

Table 6 – continued

Column Name	Data Type	Constraints	Description
application_id	UUID	FK → EOI_APPLICATION	Parent application
product_code	VARCHAR(20)	FK → FP_PRODUCT	Product being applied for
requested_coverage_amount	DECIMAL(12,2)	NOT NULL	Amount requested (e.g., 50000.00)
current_coverage_amount	DECIMAL(12,2)		Existing coverage amount (for increases)
approved_coverage_amount	DECIMAL(12,2)		Final approved amount
premium_amount	DECIMAL(10,2)		Calculated premium
product_status	VARCHAR(30)	DEFAULT 'Pending'	'Pending', 'Approved', 'Denied', 'Approved with Exclusion'
exclusion_details	TEXT		Specific exclusions if applicable
rating_class	VARCHAR(20)		'Standard', 'Rated', 'Declined'
is_guaranteed_issue	BOOLEAN		Whether this line qualified for GI
created_at	TIMESTAMP	DEFAULT NOW()	
updated_at	TIMESTAMP	DEFAULT NOW()	

Normalization: Separates product-specific details from the main application. Prevents storing variable number of products as columns in EOI_APPLICATION.

6. EOI_DOCUMENT (Document Management)

Description: Tracks all physical and digital documents associated with an EOI application.

Column Name	Data Type	Constraints	Description
document_id	UUID	PK	Unique document identifier
application_id	UUID	FK → EOI_APPLICATION	Parent application
document_type	VARCHAR(50)	NOT NULL	'EOI_Form', 'Questionnaire', 'Medical_Record', 'ID_Proof', 'Census_Sheet'

Table 7 – continued

Column Name	Data Type	Constraints	Description
document_subtype	VARCHAR(50)		‘Life_EOI’, ‘Disability_EOI’, ‘Cancer_Q’, ‘Cardiac_Q’, ‘Attend- ing_Physician_Statement’
source_channel	VARCHAR(30)	NOT NULL	‘Mail’, ‘Fax’, ‘Email_Attachment’, ‘Portal_Upload’, ‘Adobe_Sign’
file_name	VARCHAR(255)		Original file name
file_format	VARCHAR(10)		‘PDF’, ‘TIFF’, ‘PNG’, ‘JPG’, ‘DOCX’
file_size_bytes	BIGINT		Size in bytes
storage_path	VARCHAR(500)		OnBase document handle or file path
onbase_doc_id	VARCHAR(100)		OnBase document ID if integrated
received_date	TIMESTAMP	NOT NULL	When document was received
processed_date	TIMESTAMP		When document was processed/indexed
ocr_confidence_score	DECIMAL(3,2)		OCR accuracy score (0.00 - 1.00)
is_indexed	BOOLEAN	DEFAULT FALSE	Whether document has been indexed
indexed_by	VARCHAR(100)		‘SYSTEM_AUTO’ or user ID
checksum_hash	VARCHAR(64)		SHA-256 hash for integrity verification
retention_expiry_date	DATE		Date when document can be purged per retention policy
created_at	TIMESTAMP	DEFAULT NOW()	

Normalization: Documents are separate from applications to support multiple documents per application (initial form + questionnaires + medical records). Document metadata is separated from binary content (stored in OnBase/file system).

7. INDEXING_LOG (Audit Trail for Intake)

Description: Detailed audit trail of the indexing process, capturing both manual and automated indexing events.

Column Name	Data Type	Constraints	Description
log_id	UUID	PK	Unique log entry
document_id	UUID	FK → EOI_DOCUMENT	Document being indexed
application_id	UUID	FK → EOI_APPLICATION	Related application
indexing_type	VARCHAR(20)	NOT NULL	‘Manual’, ‘Automated_OCR’, ‘Automated_Barcode’, ‘API_Import’
started_at	TIMESTAMP	NOT NULL	When indexing began
completed_at	TIMESTAMP		When indexing completed
indexer_user_id	VARCHAR(100)		User ID (null for automated)
indexer_name	VARCHAR(255)		Display name of indexer
fields_extracted	JSONB		Key-value pairs of extracted fields: {“ssn”: “xxx”, “name”: “John Doe”, ...}
fields_corrected	JSONB		Fields manually corrected during QA
correction_reason	TEXT		Why corrections were needed
ocr_engine_version	VARCHAR(50)		Version of OCR software used
validation_errors	JSONB		Array of validation failures
status	VARCHAR(20)	NOT NULL	‘Success’, ‘Partial’, ‘Failed’, ‘Pending_QA’
created_at	TIMESTAMP	DEFAULT NOW()	

Normalization: Audit data is separated from the document itself. JSONB fields allow flexible schema for extracted data without altering table structure when new form versions are introduced.

8. HEALTH_QUESTIONNAIRE (Medical History)

Description: Stores responses to the core Medical History Statement (MHS) and any follow-up condition-specific questionnaires.

Column Name	Data Type	Constraints	Description
questionnaire_id	UUID	PK	Unique questionnaire instance
application_id	UUID	FK → EOI_APPLICATION	Parent application
document_id	UUID	FK → EOI_DOCUMENT	Link to the actual form document

Table 9 – continued

Column Name	Data Type	Constraints	Description
questionnaire_type	VARCHAR(50)	NOT NULL	‘Initial_MHS’, ‘Cancer’, ‘Cardiac’, ‘Diabetes’, ‘Mental_Health’, ‘Musculoskeletal’, ‘Respiratory’
questionnaire_version	VARCHAR(10)		Form version number
condition_category	VARCHAR(50)		Broad condition category
condition_specific	VARCHAR(100)		Specific condition name
diagnosis_date	DATE		When condition was diagnosed
treatment_status	VARCHAR(30)		‘Ongoing’, ‘Completed’, ‘In_Remission’, ‘Resolved’
medications	TEXT		List of current medications
physician_name	VARCHAR(255)		Treating physician name
physician_npi	VARCHAR(10)		National Provider Identifier
physician_phone	VARCHAR(20)		
physician_fax	VARCHAR(20)		
hospital_name	VARCHAR(255)		Hospital/facility name
hospital_address	TEXT		
is_prescription_pending	BOOLEAN		Whether surgery/treatment is pending
pending_procedure_date	DATE		Scheduled date if applicable
additional_notes	TEXT		Free-text medical details
is_complete	BOOLEAN	DEFAULT FALSE	Whether all required fields are filled
created_at	TIMESTAMP	DEFAULT NOW()	
updated_at	TIMESTAMP	DEFAULT NOW()	

Normalization: Medical history is separated from the main application because:

- One application can trigger multiple condition-specific questionnaires
- Medical data has different access controls (HIPAA)
- Questionnaires can be added asynchronously after initial intake

9. UNDERWRITING_REVIEW

Description: Tracks the underwriter’s review process and case management.

Column Name	Data Type	Constraints	Description
review_id	UUID	PK	Unique review record

Table 10 – continued

Column Name	Data Type	Constraints	Description
application_id	UUID	FK → EOI_APPLICATION	Parent application
underwriter_id	VARCHAR(50)	NOT NULL	Underwriter employee ID
underwriter_name	VARCHAR(255)		Underwriter display name
review_queue	VARCHAR(50)		‘Standard’, ‘Expedited’, ‘Complex’, ‘Senior_Review’
assigned_date	TIMESTAMP	NOT NULL	When case was assigned
started_date	TIMESTAMP		When underwriter opened the case
complexity_score	INT		1-10 calculated complexity rating
risk_factors	JSONB		Array of identified risk factors
required_medical_records	BOOLEAN	DEFAULT FALSE	Whether APS/medical records were requested
paramedical_required	BOOLEAN	DEFAULT FALSE	Whether paramed exam is needed
paramedical_vendor	VARCHAR(100)		Vendor name (e.g., ExamOne, APPS)
paramedical_scheduled_date	DATE		
paramedical_completed_date	DATE		
review_notes	TEXT		Underwriter’ s case notes
internal_referral_reason	VARCHAR(100)		Reason if escalated to medical director
referred_to_medical_director	BOOLEAN	DEFAULT FALSE	
created_at	TIMESTAMP	DEFAULT NOW()	
updated_at	TIMESTAMP	DEFAULT NOW()	

Normalization: Review data is separated from the application because one application may have multiple review cycles (if pended for additional info and re-submitted).

10. UNDERWRITING_DECISION

Description: Final underwriting decisions and communication tracking.

Column Name	Data Type	Constraints	Description
decision_id	UUID	PK	Unique decision record
application_id	UUID	FK → EOI_APPLICATION	Parent application
review_id	UUID	FK → UNDERWRITING_REVIEW	Related review

Table 11 – continued

Column Name	Data Type	Constraints	Description
decision_type	VARCHAR(30)	NOT NULL	‘Approved’, ‘Approved_Rated’, ‘Approved_Exclusion’, ‘Denied’, ‘Withdrawn’, ‘Pending’
decision_category	VARCHAR(30)		‘Standard’, ‘Substandard’, ‘Declined’, ‘Postponed’
decision_reason_code	VARCHAR(20)		Standardized reason code
decision_reason_description	TEXT		Detailed explanation
approved_amount	DECIMAL(12,2)		Final approved coverage amount
approved_premium_rate	DECIMAL(8,4)		Premium rate per \$1,000
exclusion_codes	JSONB		Array of exclusion condition codes
exclusion_rider_text	TEXT		Legal exclusion language
effective_date	DATE		When approved coverage becomes effective
decision_date	TIMESTAMP	NOT NULL	When decision was made
communicated_date	TIMESTAMP		When decision was sent to applicant
communication_method	VARCHAR(20)		‘Mail’, ‘Email’, ‘Portal’, ‘Phone’
employer_notified_date	TIMESTAMP		When employer was notified
appeal_deadline_date	DATE		Deadline for applicant to appeal
created_at	TIMESTAMP	DEFAULT NOW()	

Normalization: Decision is separate from review to support appeals (new decision on same application) and to maintain historical decision records if decisions are amended.

11. MEDICAL_RECORD_REQUEST

Description: Tracks requests for Attending Physician Statements (APS) and other medical records.

Column Name	Data Type	Constraints	Description
request_id	UUID	PK	Unique request ID
application_id	UUID	FK → EOI_APPLICATION	Parent application
review_id	UUID	FK → UNDERWRIT- ING_REVIEW	Related review
provider_name	VARCHAR(255)	NOT NULL	Physician/facility name

Table 12 – continued

Column Name	Data Type	Constraints	Description
provider_npi	VARCHAR(10)		
provider_address	TEXT		
provider_phone	VARCHAR(20)		
provider_fax	VARCHAR(20)		
request_type	VARCHAR(50)		‘APS’, ‘Lab_Results’, ‘Hospital_Records’, ‘Specialist_Consult’
request_date	DATE	NOT NULL	When request was sent
follow_up_date	DATE		Scheduled follow-up
received_date	DATE		When records were received
document_id	UUID	FK → EOI_DOCUMENT	Link to received document
status	VARCHAR(20)	DEFAULT ‘Pending’	‘Pending’, ‘Sent’, ‘Follow_Up’, ‘Received’, ‘Not_Available’
notes	TEXT		
created_at	TIMESTAMP	DEFAULT NOW()	
updated_at	TIMESTAMP	DEFAULT NOW()	

Normalization: Medical record requests are a separate entity because multiple providers can be contacted for one application, each with independent timelines.

12. CENSUS_BATCH (Future Scope)

Description: Batch enrollment submissions from employers (census/non-census). Tabled for future implementation.

Column Name	Data Type	Constraints	Description
batch_id	UUID	PK	Unique batch identifier
employer_id	UUID	FK → EMPLOYER	Submitting employer
batch_number	VARCHAR(50)	UNIQUE	Business-facing batch number
batch_type	VARCHAR(20)		‘Census’, ‘Non_Census’, ‘Open_Enrollment’
submission_date	DATE	NOT NULL	
total_records	INT		Number of employees in batch
processed_records	INT	DEFAULT 0	
failed_records	INT	DEFAULT 0	
file_name	VARCHAR(255)		Original upload file
file_format	VARCHAR(10)		‘CSV’, ‘XLSX’, ‘XML’, ‘EDI’

Table 13 – continued

Column Name	Data Type	Constraints	Description
validation_status	VARCHAR(20)		‘Pending’, ‘Validating’, ‘Validated’, ‘Failed’
processing_status	VARCHAR(20)	DEFAULT ‘Pending’	‘Pending’, ‘Processing’, ‘Completed’, ‘Partial’
submitted_by	VARCHAR(255)		HR contact
created_at	TIMESTAMP	DEFAULT NOW()	
completed_at	TIMESTAMP		

13. AUDIT_TRAIL (System-Wide)

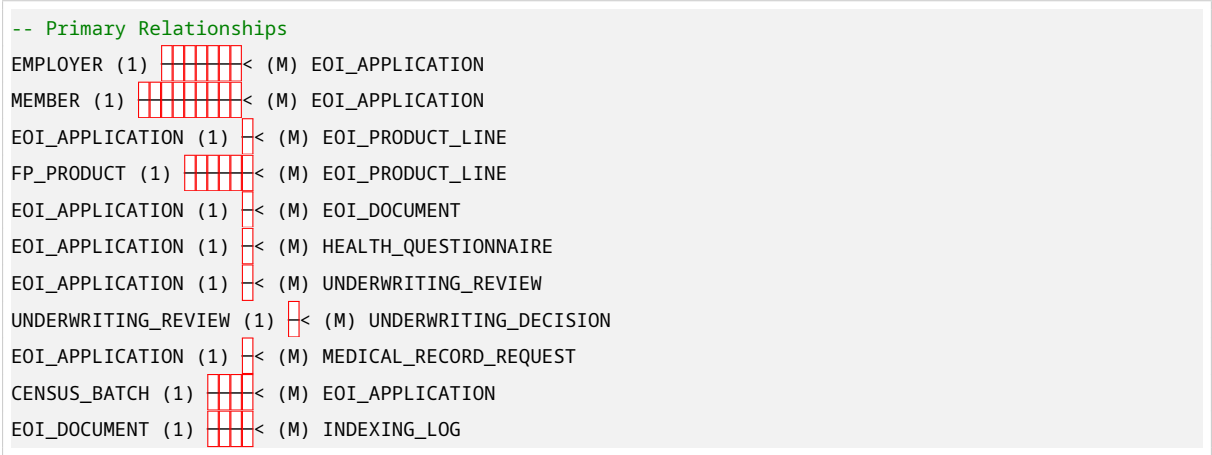
Description: Comprehensive change tracking for compliance and debugging.

Column Name	Data Type	Constraints	Description
audit_id	UUID	PK	Unique audit record
table_name	VARCHAR(50)	NOT NULL	Table being modified
record_id	UUID	NOT NULL	PK of affected record
action	VARCHAR(10)	NOT NULL	‘INSERT’, ‘UPDATE’, ‘DELETE’
old_values	JSONB		Previous state
new_values	JSONB		New state
changed_by	VARCHAR(100)		User ID or ‘SYSTEM’
changed_at	TIMESTAMP	DEFAULT NOW()	
session_id	VARCHAR(100)		Application session
ip_address	VARCHAR(45)		Source IP

Normalization Summary

Normal Form	Compliance
1NF	✔ All columns atomic. No repeating groups. Array data stored in junction tables or JSONB only for truly variable attributes (extracted fields).
2NF	✔ All non-key attributes fully dependent on the entire PK. No partial dependencies.
3NF	✔ No transitive dependencies. Employer details in EMPLOYER, member details in MEMBER, product definitions in FP_PRODUCT.
Referential Integrity	✔ Foreign keys enforce relationships. Cascading rules defined for soft deletes.

Key Relationships



Indexing Strategy

Table	Index Columns	Purpose
EOI_APPLICATION	(application_number)	Unique lookup
EOI_APPLICATION	(employer_id, application_date)	Employer reporting
EOI_APPLICATION	(member_id, application_date)	Member history
EOI_APPLICATION	(application_status, intake_channel)	Workflow queue
EOI_APPLICATION	(adobe_agreement_id)	Adobe Sign reconciliation
MEMBER	(ssn)	Encrypted, but indexed for lookup
MEMBER	(last_name, first_name, date_of_birth)	Name search
EOI_DOCUMENT	(application_id, document_type)	Document retrieval
EOI_DOCUMENT	(onbase_doc_id)	OnBase cross-reference
INDEXING_LOG	(document_id, status)	Processing queue
HEALTH_QUESTIONNAIRE	(application_id, questionnaire_type)	Medical review
UNDERWRITING_REVIEW	(underwriter_id, assigned_date)	Workload management
UNDERWRITING_DECISION	(application_id, decision_date)	Decision history

Data Retention & Compliance Notes

1. **PHI/PII:** SSN and medical history encrypted at rest (AES-256). Field-level encryption recommended for SSN.

2. **HIPAA:** HEALTH_QUESTIONNAIRE and UNDERWRITING_REVIEW access logged in AUDIT_TRAIL.

3. **Retention:** EOI documents retained per state regulations (typically 7 years). Retention expiry tracked in EOI_DOCUMENT.

4. **Audit:** All status changes on EOI_APPLICATION logged in AUDIT_TRAIL.

5. **Adobe e-sign:** Agreement IDs preserved for legal non-repudiation.

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